# CIS 6270 Practice Exam 1 for Exam 2 SOLUTIONS

Professor Pranam Chatterjee
TAs: Sophie Vincoff, Tong Chen, and Rosie Zhang

Fall 2026

Follow these instructions for a timed practice attempt:

- Please write your name on each page. Allow yourself 90 minutes.
- Complete your first attempt individually before consulting the solutions. Record questions to discuss afterward with the instructor and teaching assistants.
- Calculators are not permitted. Every part is multiple choice with six options. Work in the space provided, then box exactly one option letter for every part. When grading, count only the boxed letter.
- Each labeled part is worth 0.5 points and receives either 0 or 0.5 points. There is no partial credit.
- You may use a one-page cheat sheet (8.5in $\times$ 11in, one side only) containing only material provided in this class. Do not use a calculator, phone, or other device during the timed attempt. Keep your signed cheat sheet with your completed work.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I will use this practice exam to assess my own understanding. I will complete the timed attempt using my own work before consulting the solutions or discussing answers.

This practice exam is worth 12 points and does not contribute to your course grade.

Name: _____

Signature: _____

*This page is left intentionally blank for scratch work*

## This practice exam is worth 12 points.

### Beating Autoregression on LM1B with Staggered Block Diffusion

You lead a group that trains language models. As you know from Lecture 4, an autoregressive transformer still has the lowest perplexity among the models we studied on LM1B, a benchmark built from English news sentences. With 327B training tokens, a retrained transformer had a test perplexity of 20.86, and MDLM had a perplexity bound of 23.00. In the BD3-LM comparison, which used the same training budget for every model, the transformer had a test perplexity of 22.83, while MDLM and BD3-LM with blocks of four had bounds of 31.78 and 28.23. As its blocks shrink, BD3-LM approaches autoregression, yet on LM1B, every discrete diffusion model in the lecture fell short of the transformer.

In block diffusion, we keep the chain rule between blocks and fill each block by masked diffusion. Blocks of one token recover autoregression, but then we generate only one token per network call. In this practice exam, you will design a new block diffusion model, which we call staggered block diffusion. We keep the blocks of four from BD3-LM and mask later positions sooner than earlier ones. During generation, we therefore tend to fill each block from left to right. During training, we still corrupt every block at random noise levels.

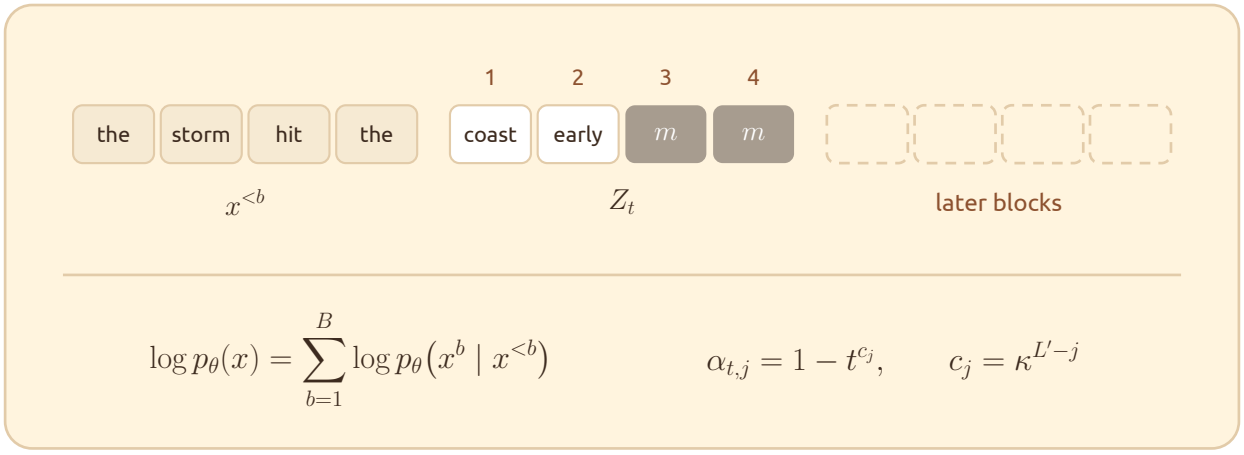



**Figure 1.** Staggered block diffusion on LM1B. We split each sentence into blocks of $L' = 4$ tokens and generate the blocks in order, each given the clean blocks before it. Inside the current block, offset $j$ keeps its token with probability $\alpha_{t,j} = 1 - t^{c_j}$, with larger exponents $c_j$ at earlier offsets. Corruption runs from the clean block at $t = 0$ to the all-mask block at $t = 1$, and generation runs back from $t = 1$ to $t = 0$.

Figure 1 outlines the construction. You will follow the steps from the end of Lecture 4 for building a new discrete diffusion model. You will choose its rate matrix, compute its forward marginal, derive its reverse rates and loss, sample from it, and steer it toward a topic with Doob's $h$-transform. You will then interpret hypothetical LM1B results for this model.

## Research brief and notation

Each training sentence $x = (x_1, \ldots, x_L)$ has $L = 128$ tokens from the LM1B vocabulary $\mathcal{V}$, which contains $K = 30{,}522$ tokens, and $m \notin \mathcal{V}$ denotes the mask. We split $x$ into $B = L/L'$ blocks of $L'$ tokens. We write $x^b$ for block $b$, $x^{<b}$ for the clean blocks before it, and $x^b_j$ for the token at offset $j \in \{1, \ldots, L'\}$ of block $b$. As in BD3-LM, the chain rule holds between blocks:

$$\log p_\theta(x) = \sum_{b=1}^{B} \log p_\theta\big(x^b \mid x^{<b}\big).$$

We model each factor with masked diffusion on the current block, given the clean prefix. Time $t$ runs from $t = 0$, where the block is clean, to $t = 1$, where every offset is masked. The noisy block is $Z_t = (Z_{t,1}, \ldots, Z_{t,L'})$, and $z$ denotes one of its values. Every offset keeps its token independently, with survival probability

$$\alpha_{t,j} = 1 - t^{c_j}, \qquad c_j = \kappa^{L'-j}, \qquad \kappa \geq 1.$$

With $\kappa = 1$, every offset follows MDLM's linear schedule $\alpha_t = 1 - t$, which recovers BD3-LM with that schedule. With $\kappa > 1$, earlier offsets survive longer. Our main runs use $L' = 4$ and $\kappa = 2$, so $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$.

The denoiser $d^j_\theta(a \mid z, x^{<b})$ is the probability that offset $j$ of the current block holds the clean token $a \in \mathcal{V}$, given the noisy block $z$ and the clean prefix. We write $z^{j \leftarrow a}$ for $z$ with offset $j$ set to $a$. All rates, marginals, and posteriors below condition on the prefix $x^{<b}$. Perplexity is the exponential of the average negative log-likelihood per token in nats, and log denotes the natural logarithm.

| **Step** | **Points** |
|---|---|
| 1. Locate the gap to autoregression | 1.5 |
| 2. Choose the staggered rate matrix | 2.0 |
| 3. Derive the reverse rates and the loss | 2.5 |
| 4. Sample a block | 2.0 |
| 5. Guide generation toward a topic | 2.0 |
| 6. Interpret hypothetical LM1B results | 2.0 |
| **Total: 24 parts** | **12.0** |

## Step 1: Locating the Gap to Autoregression (1.5 points)

Before we change the model, we need to read the LM1B numbers correctly and connect the block factorization to autoregression.

**(a)** We know only bounds on the perplexity of the diffusion models, while we know the transformer's perplexity exactly. We should first check what a bound allows us to conclude.

In the BD3-LM comparison, the transformer had a test perplexity of 22.83, BD3-LM with blocks of four had a bound of 28.23, and MDLM had a bound of 31.78. Which conclusion do these three numbers support? Choose ONE.

A. BD3-LM's test perplexity exceeds the transformer's by exactly 5.40.

B. BD3-LM's test perplexity is at least 28.23, so it is worse than the transformer's.

C. BD3-LM's test perplexity is at most 28.23, so it may lie above or below 22.83.

D. MDLM's test perplexity exceeds BD3-LM's, since 31.78 is larger than 28.23.

E. BD3-LM's test perplexity lies somewhere between 22.83 and 28.23.

F. The transformer's 22.83 is also a bound, so no ranking of the three models follows.

(0.5 points)

Every diffusion number in this comparison bounds the negative log-likelihood, since $-\log p_\theta(x) \le \mathcal{L}(x)$ for each test sentence. Averaging over the test tokens and exponentiating preserve the inequality, because the exponential is increasing:

$$\text{test perplexity} = \exp\Big(\tfrac{1}{N}\textstyle\sum(-\log p_\theta)\Big) \le \exp\Big(\tfrac{1}{N}\textstyle\sum \mathcal{L}\Big) = \text{reported bound.}$$

So the test perplexity of BD3-LM is at most 28.23. The transformer's chain rule gives its likelihood exactly, so 22.83 is its exact perplexity. In Lecture 4, we noted that a high bound may be loose, so the true value for BD3-LM can lie anywhere up to 28.23, including below 22.83. Therefore, Option C.

Option A equates the bound with the exact perplexity. Option B reverses the inequality. Option D compares two upper bounds, which cannot order the true values. Option E adds a lower limit of 22.83 that nothing in the comparison establishes. Option F misstates the transformer's number, which is exact.

**(b)** By the chain rule, the probability of a sentence is a product of block probabilities. We need to convert that product into the perplexity that we report on LM1B.

A model splits a sentence of six tokens into three blocks of two tokens and assigns the blocks probabilities 1/2, 1/4, and 1/8, each given its prefix. What is the perplexity of this sentence? Choose ONE.

A. 2

B. 4

C. 8

D. 64

E. 14/3

F. $\sqrt{2}$

(0.5 points)

By the chain rule, the sentence probability is the product of the block probabilities:

$$p_\theta(x) = \tfrac{1}{2} \cdot \tfrac{1}{4} \cdot \tfrac{1}{8} = \tfrac{1}{64} = 2^{-6}.$$

The negative log-likelihood is $\log 64 = 6 \log 2$ nats. To compute perplexity, we divide by the six tokens and exponentiate:

$$\exp\left(\frac{6 \log 2}{6}\right) = 2.$$

Therefore, Option A. On average, the model was as uncertain as a fair choice between two tokens at each position.

Option B averages over the three blocks and gives $64^{1/3} = 4$. Option C divides by the block size of two and gives $64^{1/2} = 8$. Option D does not average at all. Option E averages the inverse block probabilities, $(2 + 4 + 8)/3$. Option F divides by twelve and gives $64^{1/12}$.

**(c)** In Lecture 4, we noted that blocks of one token recover autoregression. We can check this directly from the block loss.

Take $L' = 1$ with MDLM's linear schedule $\alpha_t = 1 - t$, so every block contains one token $x^b$ and its loss is

$$\int_0^1 \frac{1}{t} \Pr(Z_t = m) \left(-\log d_\theta(x^b \mid m, x^{<b})\right) \mathrm{d}t.$$

The denoiser does not receive $t$, so its prediction is the same at every time when the token is masked. What does this loss equal? Choose ONE.

A. $\frac{1}{2}\left(-\log d_\theta(x^b \mid m, x^{<b})\right)$

B. $2\left(-\log d_\theta(x^b \mid m, x^{<b})\right)$

C. It diverges, because $\int_0^1 \mathrm{d}t/t$ is infinite.

D. $-\log d_\theta(x^b \mid m, x^{<b}) + \log K$

E. $-\log d_\theta(x^b \mid m, x^{<b})$

F. $-\log d_\theta(x^b \mid m, x^{<b}) - 1 + d_\theta(x^b \mid m, x^{<b})$

(0.5 points)

With $\alpha_t = 1-t$, the single token is masked with probability $\Pr(Z_t = m) = 1-\alpha_t = t$. The denoiser does not receive $t$, so $-\log d_\theta(x^b \mid m, x^{<b})$ is a constant, and the integrand is constant in time:

$$\int_0^1 \frac{1}{t} \cdot t \cdot \left(-\log d_\theta(x^b \mid m, x^{<b})\right) \mathrm{d}t = -\log d_\theta(x^b \mid m, x^{<b}).$$

When the block is masked, the denoiser receives only the prefix, so $d_\theta(x^b \mid m, x^{<b})$ predicts the next token from the tokens before it. Each block loss is the autoregressive cross-entropy of one token, and the sum over blocks is the transformer's loss. Therefore, Option E.

Option A uses the weight $-\dot{\alpha}_t = 1$ without dividing by the masking probability, which gives $\int_0^1 t\,\mathrm{d}t = \frac{1}{2}$. Option B doubles the correct value with no basis in the integral. Option C omits the masking probability $t$, which cancels the factor $1/t$. Option D adds a prior term, which is zero because the process ends at the mask and generation starts there. Option F keeps the terms $-A + B$ of the rate error $\ell(A, B)$ for the correct token alone. Summing $B$ over every token gives $\sum_a d_\theta(a) = 1$, which cancels $-A$ exactly.

## Step 2: Choosing the Staggered Rate Matrix (2 points)

In GenMD4, each token had its own masking rate. We can similarly assign a rate to each offset so that later offsets are masked sooner. We then need to confirm that these rates satisfy the first three facts from Lecture 4.

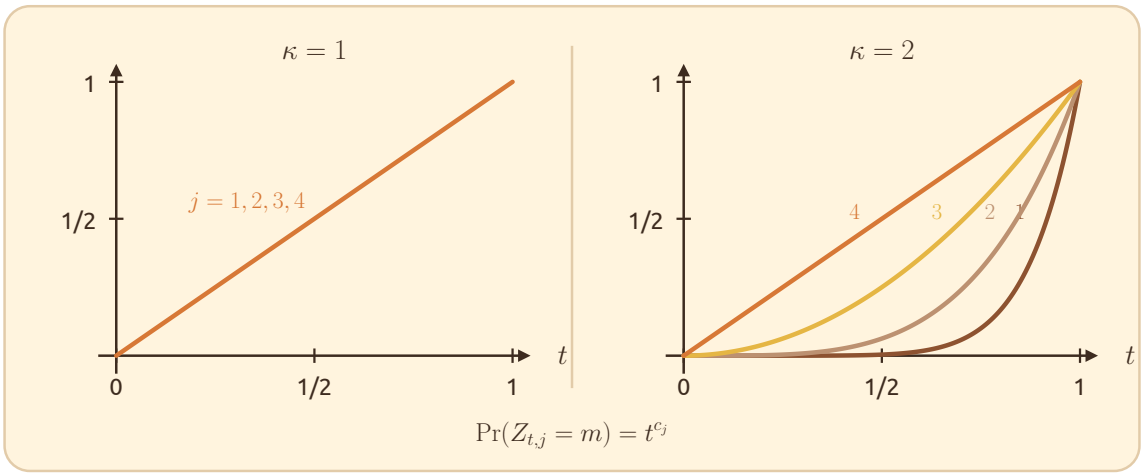



**Figure 2.** Masking probability $\Pr(Z_{t,j} = m) = t^{c_j}$ at each offset $j$ of a block of $L' = 4$. With $\kappa = 1$, all four offsets follow the linear schedule. With $\kappa = 2$, the exponents are $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$, and offset 4 is masked soonest.

**(a)** To write the rate matrix, we first need the masking rate corresponding to our survival schedule.

A clean token at offset $j$ jumps to $m$ at rate $\beta_j(t)$, and $m$ never jumps back. Which rate $\beta_j(t)$ produces the survival schedule $\alpha_{t,j} = 1 - t^{c_j}$? Choose ONE.

A. $\dfrac{c_j}{t}$

B. $-\log\left(1 - t^{c_j}\right)$

C. $c_j\, t^{c_j - 1}$

D. $\dfrac{c_j\, t^{c_j - 1}}{1 - t^{c_j}}$

E. $\dfrac{c_j\, t^{c_j}}{1 - t^{c_j}}$

F. $\dfrac{c_j}{1 - t}$

(0.5 points)

In Lecture 4, we related the survival probability to the clock through $\alpha = e^{-\tau}$, with $\tau$ growing at

rate $\beta$, so $\beta = -\dot{\alpha}/\alpha$. Differentiate the schedule:

$$\dot{\alpha}_{t,j} = \frac{\mathrm{d}}{\mathrm{d}t}\big(1 - t^{c_j}\big) = -c_j\, t^{c_j - 1}.$$

Divide by the survival probability:

$$\beta_j(t) = -\frac{\dot{\alpha}_{t,j}}{\alpha_{t,j}} = \frac{c_j\, t^{c_j-1}}{1 - t^{c_j}}.$$

Therefore, Option D. In the row of the rate matrix for a clean token $a$ at offset $j$, the rate toward $m$ is $\beta_j(t) \geq 0$, the diagonal entry is $-\beta_j(t)$, and every other entry is zero. The row of $m$ is zero. Every off-diagonal rate is nonnegative and every row sums to zero, so the first fact holds. With $c_j = 1$, the rate is $1/(1-t)$, the rate of MDLM's linear schedule.

Option A is the reveal weight $c_j/t$ of the reverse process, which divides by $1 - \alpha_{t,j}$. Option B is the clock $\tau_j(t)$ itself. Option C is $-\dot{\alpha}_{t,j}$ without the division by $\alpha_{t,j}$. Option E differentiates $t^{c_j}$ incorrectly. Option F agrees with the correct rate only when $c_j = 1$.

**(b)** Using this rate, we can track how probability flows into the mask at one offset.

Let $p_t(m) = \Pr(Z_{t,j} = m)$. Which equation is the master equation at the mask state? Choose ONE.

A. $\dot{p}_t(m) = \beta_j(t)\, p_t(m)$

B. $\dot{p}_t(m) = \beta_j(t)\big(1 - p_t(m)\big)$

C. $\dot{p}_t(m) = \beta_j(t)\big(1 - p_t(m)\big) - \beta_j(t)\, p_t(m)$

D. $\dot{p}_t(m) = -\beta_j(t)\big(1 - p_t(m)\big)$

E. $\dot{p}_t(m) = \dfrac{c_j}{t}\big(1 - p_t(m)\big)$

F. $\dot{p}_t(m) = \beta_j(t)$

(0.5 points)

Flux arrives at $m$ from every clean token $a$ at rate $\beta_j(t)$. All probability outside $m$ is assigned to clean tokens, so the incoming flux is

$$\sum_{a \in \mathcal{V}} p_t(a)\, \beta_j(t) = \beta_j(t)\big(1 - p_t(m)\big).$$

Because the mask is absorbing, there is no outgoing flux from $m$. The master equation at the mask is

$$\dot{p}_t(m) = \beta_j(t)\big(1 - p_t(m)\big).$$

Therefore, Option B. The marginal $p_t(m) = t^{c_j}$ satisfies it, since $\beta_j(t)\big(1 - t^{c_j}\big) = c_j\, t^{c_j - 1}$, which is the derivative of $t^{c_j}$.

Option A weights the rate by the probability of the mask itself, as if flux left $m$. Option C subtracts a flux out of $m$ that does not exist, since $R_t(m, a) = 0$. Option D has the wrong sign, so $p_t(m)$ would fall. Option E puts the reverse weight $c_j/t$ into the forward equation. Option F omits the probability of the clean tokens that the flux leaves.

**(c)** We can also solve the master equation in closed form, which gives the second fact.

At offset $j$, the rate matrix is $R_t = \beta_j(t) R_\pi$ with $R_\pi = \mathbf{1}\pi^\top - I$ and $\pi = \delta_m$. Define the clock $\tau_j(t) = \int_0^t \beta_j(v)\,\mathrm{d}v$. Which transition matrix $P_{0,t}$ gives the forward marginal of offset $j$? Here $u$ denotes the uniform distribution over $\mathcal{V}$. Choose ONE.

A. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = -\log\big(1 - t^{c_j}\big)$

B. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = c_j\, t$

C. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}u^\top$ with $\tau_j(t) = c_j\, t$

D. $\big(1 - e^{-\tau_j(t)}\big) I + e^{-\tau_j(t)}\mathbf{1}\delta_m^\top$ with $\tau_j(t) = c_j\, t$

E. $e^{\tau_j(t)} I + \big(1 - e^{\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = -\log\big(1 - t^{c_j}\big)$

F. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = t^{c_j}$

(0.5 points)

With $\pi = \delta_m$, every row of $M = \mathbf{1}\pi^\top$ equals $\delta_m$, and $M^2 = M$ because $\pi^\top\mathbf{1} = 1$. In Lecture 4, we used this idempotence to sum the exponential series:

$$e^{\tau R_\pi} = e^{-\tau} I + \big(1 - e^{-\tau}\big)\mathbf{1}\pi^\top.$$

Our rate matrix $\beta_j(t) R_\pi$ commutes with itself at every time, so $P_{0,t} = e^{\tau_j(t) R_\pi}$ with the clock

$$\tau_j(t) = \int_0^t \frac{c_j\, v^{c_j - 1}}{1 - v^{c_j}}\,\mathrm{d}v = -\log\big(1 - t^{c_j}\big).$$

Then $e^{-\tau_j(t)} = 1 - t^{c_j} = \alpha_{t,j}$, so a clean token stays with probability $\alpha_{t,j}$ and becomes $m$ with probability $t^{c_j}$, as the schedule requires. Therefore, $\boxed{\text{Option A.}}$ As $t \to 1$, $e^{-\tau_j(t)} \to 0$, so every row approaches $\delta_m$ and the block ends at all masks. Since $\delta_m R_\pi = \delta_m M - \delta_m = 0$, the all-mask state is stationary, satisfying the third fact. We start generation from this state.

Option B uses the constant rate $c_j$, so a fraction $e^{-c_j} > 0$ of tokens survives at $t = 1$ and the block never reaches all masks. Option C also uses the constant rate, and it spreads the lost probability uniformly over $\mathcal{V}$, which is the uniform-noise kernel of UDLM. Option D also uses the constant rate, and it swaps the survival and masking probabilities, so the block would start fully masked at $t = 0$. Option E flips the sign of the clock, which gives entries above one. Option F omits the logarithm from the clock, so a fraction $e^{-1}$ of tokens survives at $t = 1$.

**(d)** In Step 4, we will need the probability that a visible token becomes masked between two times. Survival to a later time requires survival to the earlier time first.

For $s < t$, what is the probability that offset $j$ is masked at time $t$, given that it holds its clean token $a$ at time $s$? Choose ONE.

A. $\dfrac{t^{c_j} - s^{c_j}}{1 - s^{c_j}}$

B. $\dfrac{t^{c_j} - s^{c_j}}{t^{c_j}}$

C. $\dfrac{1 - t^{c_j}}{1 - s^{c_j}}$

D. $t^{c_j} - s^{c_j}$

E. $1 - e^{-c_j(t-s)}$

F. $(t-s)^{c_j}$

(0.5 points)

To be visible at $t$, the token must be visible at $s$ and then survive the interval, so

$$\alpha_{t,j} = \alpha_{s,j} \Pr\big(Z_{t,j} = a \mid Z_{s,j} = a\big).$$

The interval survival probability is therefore $\alpha_{t,j}/\alpha_{s,j}$, and its complement is the probability of masking during the interval:

$$\Pr\big(Z_{t,j} = m \mid Z_{s,j} = a\big) = 1 - \frac{1 - t^{c_j}}{1 - s^{c_j}} = \frac{t^{c_j} - s^{c_j}}{1 - s^{c_j}}.$$

Therefore, Option A. If we discretize time as in D3PM, this is the masking probability of one step, and the survival probabilities of the steps multiply to $\alpha_{t,j}$.

Option B is the reverse probability of revealing a masked offset over the same interval, which we will use in Step 4. Option C is the probability of surviving the interval. Option D is the unconditional probability of masking between $s$ and $t$, without conditioning on survival to $s$. Option E uses the constant rate $c_j$. Option F does not follow from the schedule.

## Step 3: Deriving the Reverse Rates and the Loss (2.5 points)

The fourth and fifth facts concern generation and training. We need reverse rates that come from the posterior over clean tokens, and a loss that bounds the likelihood and is minimized at the true reverse rates.

**(a)** At a masked offset, we need to know how fast to reveal a token and which token to reveal. In Lecture 4, we reversed a jump with a ratio of probabilities and then averaged over the clean token with the denoiser.

Write $d$ for $d_\theta^j(a \mid z, x^{<b})$. Which pair gives the conditional reverse rate $\hat{R}_t(m \to a \mid x^b)$ at a masked offset $j$ and the model rate $\hat{R}_t^\theta(m \to a)$? Choose ONE.

A. $\big(\beta_j(t)\,\delta(a, x_j^b),\quad \beta_j(t)\,d\big)$

B. $\big(\frac{c_j}{t}\,\delta(a, x_j^b),\quad \beta_j(t)\,d\big)$

C. $\big(\frac{1}{t}\,\delta(a, x_j^b),\quad \frac{1}{t}\,d\big)$

D. $\big(c_j\,t^{c_j-1}\,\delta(a, x_j^b),\quad c_j\,t^{c_j-1}\,d\big)$

E. $\big(\frac{c_j}{t}\,\delta(a, x_j^b),\quad \frac{c_j}{t}\,d\big)$

F. $\big(\beta_j(t)\,\delta(a, x_j^b),\quad \frac{c_j}{t}\,d\big)$

(0.5 points)

In Lecture 4, we reversed a jump by multiplying the forward rate by a probability ratio. Given the clean block, the jump $m \to a$ at offset $j$ reverses the forward jump $a \to m$, whose rate is $\beta_j(t)$:

$$\hat{R}_t(m \to a \mid x^b) = \beta_j(t)\,\frac{q_t(a \mid x_j^b)}{q_t(m \mid x_j^b)} = \beta_j(t)\,\frac{\alpha_{t,j}\,\delta(a, x_j^b)}{1-\alpha_{t,j}}.$$

Substitute $\beta_j(t)$ from 2(a):

$$\frac{c_j\,t^{c_j-1}}{1-t^{c_j}} \cdot \frac{\big(1-t^{c_j}\big)\,\delta(a, x_j^b)}{t^{c_j}} = \frac{c_j}{t}\,\delta(a, x_j^b).$$

The factor $c_j/t$ equals $-\dot{\alpha}_{t,j}/(1-\alpha_{t,j})$, the reveal weight $w_j(t)$, just as MDLM's weight was $-\dot{\alpha}_t/(1-\alpha_t)$. Averaging the clean token over the denoiser gives the model rate $\frac{c_j}{t}\,d$. Therefore, **Option E.** With $\kappa = 1$, every $c_j = 1$ and both rates reduce to MDLM's $1/t$. With $\kappa = 2$ at $t = 1/2$, offset 1 is revealed at total rate 16 and offset 4 at total rate 2, so early offsets fill in first.

Option A omits the probability ratio. Options B and F put the forward rate on one side of the pair only. Option C is MDLM's weight, which holds only when $c_j = 1$. Option D omits the division by the masking probability $1 - \alpha_{t,j} = t^{c_j}$.

**(b)** To train the denoiser, we compare these model rates with the conditional targets through the rate error $\ell(A, B) = A \log(A/B) - A + B$.

Sum the rate error over $a$, then over offsets and blocks. The expectation is over sentences $x$ and noisy blocks $Z_t$ drawn from their forward marginals. Here $Z_t^{<b}$ denotes the earlier blocks corrupted to the same time. Which expression is the staggered block loss? Choose ONE.

A. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$$

B. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, Z_t^{<b})\big)\Big]\, \mathrm{d}t$$

C. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b}) - 1 + d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$$

D. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b}) - 1 + d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$$

E. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$$

F. $$\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, Z_t^{<b})\big)\Big]\, \mathrm{d}t$$

(0.5 points)

At a masked offset $j$, the target and model rates toward $a$ are $A_a = \frac{c_j}{t}\,\delta(a, x_j^b)$ and $B_a = \frac{c_j}{t}\, d_\theta^j(a \mid Z_t, x^{<b})$. Sum the rate error over $a$:

$$\sum_a \ell(A_a, B_a) = \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big) - \frac{c_j}{t} + \frac{c_j}{t} \sum_{a \in \mathcal{V}} d_\theta^j(a \mid Z_t, x^{<b}).$$

The denoiser assigns zero probability to $m$. Its probabilities over $\mathcal{V}$ therefore sum to one, so the last two terms cancel. The forward process never jumps into a clean token. At a visible offset, all target and model reverse rates are therefore zero, so the offset does not contribute to the loss. Adding the masked offsets, integrating over time, and summing the blocks with the chain rule gives

$$\mathcal{L} = \sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[ \sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t.$$

Therefore, Option E. This is the BD3-LM loss with a weight $c_j/t$ that depends on the offset. In code, we mask offset $j$ with probability $t^{c_j}$ and multiply its cross-entropy by $c_j/t$.

Option A keeps MDLM's weight $1/t$, which matches only for $\kappa = 1$. Option B keeps MDLM's weight and also corrupts the prefix, although every factor of the chain rule conditions on clean earlier blocks. Option C keeps the terms $-A + B$ of the rate error for the correct token alone, although summing $B$ over every token cancels $-A$. Option D combines the errors of Options A

and C. Option F corrupts the prefix.

**(c)** In SEDD, the network predicts the probability ratios in the reverse rate directly. We should check how these ratios relate to our denoiser.

At a masked offset $j$, we wrote the ratio in Lecture 4 as

$$\frac{p_t(z^{j\leftarrow a})}{p_t(z)} = \sum_{y\in\mathcal{V}} \Pr\big(x_j^b = y \mid Z_t = z\big)\, \frac{q_t(a \mid y)}{q_t(m \mid y)},$$

where $q_t(\cdot \mid y)$ is the forward marginal of offset $j$ with clean token $y$. Write $d^j(a) = \Pr(x_j^b = a \mid Z_t = z)$ for the exact posterior. Which expression does this ratio equal for our schedule? Choose ONE.

A. $\dfrac{t^{c_j}}{1-t^{c_j}}\, d^j(a)$

B. $\dfrac{c_j}{t}\, d^j(a)$

C. $d^j(a)$

D. $\beta_j(t)\, d^j(a)$

E. $\big(1-t^{c_j}\big)\, d^j(a)$

F. $\dfrac{1-t^{c_j}}{t^{c_j}}\, d^j(a)$

(0.5 points)

For a clean token $a \in \mathcal{V}$, the forward marginal of offset $j$ gives $q_t(a \mid y) = \alpha_{t,j}\, \delta(a,y)$, and $q_t(m \mid y) = 1-\alpha_{t,j}$ for every clean token $y$. Only the term with $y = a$ is nonzero:

$$\frac{p_t(z^{j\leftarrow a})}{p_t(z)} = \sum_{y\in\mathcal{V}} \Pr\big(x_j^b = y \mid Z_t = z\big)\, \frac{\alpha_{t,j}\, \delta(a,y)}{1-\alpha_{t,j}} = \frac{1-t^{c_j}}{t^{c_j}}\, d^j(a).$$

Therefore, Option F. Multiplying this ratio by the forward rate $\beta_j(t)$ of the jump $z^{j\leftarrow a} \to z$ returns the reverse rate of 3(a):

$$\frac{c_j\, t^{c_j-1}}{1-t^{c_j}} \cdot \frac{1-t^{c_j}}{t^{c_j}}\, d^j(a) = \frac{c_j}{t}\, d^j(a).$$

A SEDD network at a masked offset would have to learn the schedule factor $(1-t^{c_j})/t^{c_j}$, which grows without limit as $t \to 0$. In our parameterization, we include that factor explicitly, so the network predicts only a distribution over $\mathcal{V}$.

Option A inverts the schedule factor. Option B is the reverse rate, the ratio multiplied by $\beta_j(t)$. Option C drops the schedule factor. Option D multiplies by the forward rate. Option E omits the division by the masking probability.

**(d)** During training, the integrand of the loss in (b) sums the weighted cross-entropies of the masked offsets.

At time $t = 1/2$, offsets 2 and 4 of a block are masked, and offsets 1 and 3 are visible. The denoiser's cross-entropies at the clean tokens are 1.0 at offset 2 and 0.5 at offset 4. With $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$, what is the contribution of these two offsets to the loss integrand? Choose ONE.

A. 3

B. 9/2

C. 9

D. 12

E. 1

F. 23/15

(0.5 points)

Each masked offset adds $\frac{c_j}{t}\left(-\log d_\theta^j\right)$ to the integrand. At $t = 1/2$, the weights are $c_2/t = 8$ and $c_4/t = 2$, so

$$8 \cdot 1.0 + 2 \cdot 0.5 = 9.$$

Therefore, Option C. At this time, an error at offset 2 counts four times as much as the same error at offset 4, because offset 2 is revealed four times as fast.

Option A uses MDLM's weight $1/t = 2$ at both offsets. Option B drops the division by $t$. Option D assigns the exponents in reverse order, as if $c_2 = 2$ and $c_4 = 8$. Option E multiplies each weight again by the masking probability $t^{c_j}$, which the sampled masks already account for. Option F uses the forward rates $\beta_2(1/2) = 8/15$ and $\beta_4(1/2) = 2$ as the weights.

**(e)** Many clean sentences are consistent with the same noisy block, so the conditional targets in (b) change from one training example to the next.

Fix the noisy block $z$ and the prefix $x^{<b}$. Over the clean blocks consistent with $z$, the conditional target rate toward $a$ at a masked offset $j$ is $A = \frac{c_j}{t}\,\delta(a, x_j^b)$. Which model rate $B$ minimizes the expected rate error $\mathbb{E}[\ell(A, B)]$ over the training sentences with prefix $x^{<b}$ whose current block is corrupted to $z$? Choose ONE.

A. $\frac{c_j}{t}\,\delta(a, \hat{x}_j)$, where $\hat{x}_j$ is the most probable clean token at offset $j$

B. $\frac{c_j}{t}\,\Pr(x_j^b = a \mid x^{<b})$, the probability given the prefix alone

C. $\beta_j(t)\,\Pr(x_j^b = a \mid x^{<b})$

D. $\frac{c_j}{t}\,\Pr(x_j^b = a \mid Z_t = z, x^{<b})$

E. No single rate minimizes it, because $A$ changes with the clean block.

F. $\frac{c_j}{t}\left(1 - \Pr(x_j^b = a \mid Z_t = z, x^{<b})\right)$

(0.5 points)

In Lecture 4, we minimized the expected rate error by setting its derivative in $B$ to zero:

$$\frac{\partial}{\partial B}\,\mathbb{E}\big[A\log(A/B) - A + B\big] = 1 - \frac{\mathbb{E}[A]}{B} = 0 \quad\Longrightarrow\quad B^\star = \mathbb{E}[A].$$

Here $A = \frac{c_j}{t}\,\delta(a, x_j^b)$, and its average over these training sentences is the posterior probability of $a$ at offset $j$:

$$B^\star = \frac{c_j}{t}\,\Pr\big(x_j^b = a \mid Z_t = z, x^{<b}\big).$$

Therefore, Option D. This is the marginal reverse rate. By the conditional rate-matching theorem, when we train on conditional targets, the optimal denoiser is this posterior.

Option A minimizes a 0–1 error, which picks the most probable token. Option B drops the visible offsets of the block, which the posterior conditions on. Option C uses the forward rate and drops the visible offsets of the block. Option E is false, because the expected rate error is convex in $B$ with the unique minimizer $\mathbb{E}[A]$. Option F uses the complement of the posterior.

## Step 4: Sampling a Block (2 points)

With a trained denoiser, we generate the blocks in order. We initialize each block with all masks and run its reverse process from $t = 1$ to $t = 0$, conditioned on the clean prefix. We then append the completed block to the prefix (Figure 3).

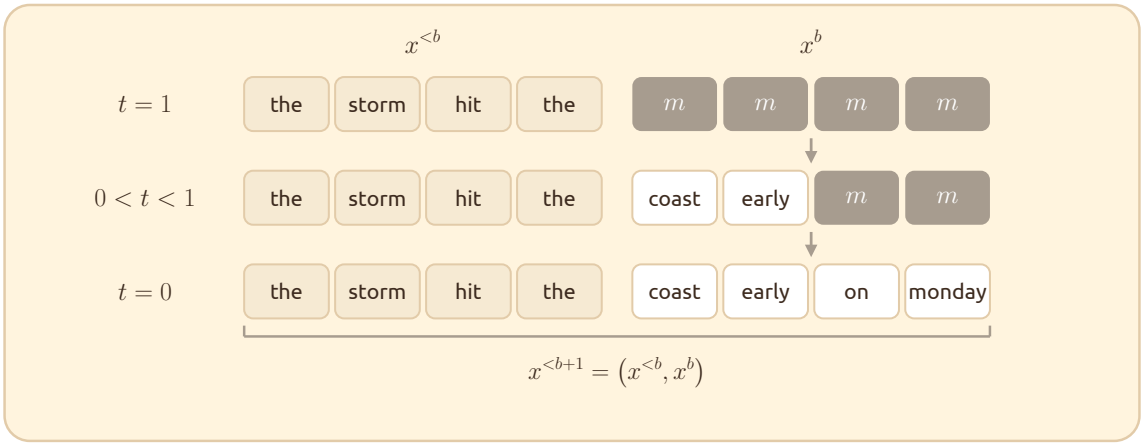



**Figure 3.** Generating one block of $L' = 4$ tokens. Given the clean prefix $x^{<b}$, the current block starts from all masks at $t = 1$ and is clean at $t = 0$. A reverse step can reveal several offsets, and a revealed token stays fixed. The completed block then joins the prefix of block $b + 1$.

**(a)** Our first reverse step starts from the all-mask block, and each offset becomes visible independently.

Take one exact posterior step from $t = 1$ to $s = 1/2$ with $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$. Which vector lists the probabilities that offsets 1, 2, 3, and 4 become visible in this step? Choose ONE.

A. $(1/2,\ 3/4,\ 15/16,\ 255/256)$

B. $(255/256,\ 15/16,\ 3/4,\ 1/2)$

C. $(1/256,\ 1/16,\ 1/4,\ 1/2)$

D. $(1/2,\ 1/2,\ 1/2,\ 1/2)$

E. $(1,\ 1,\ 1,\ 1/2)$

F. $(8/15,\ 4/15,\ 2/15,\ 1/15)$

(0.5 points)

In an exact posterior step from $t$ to $s$, a masked offset becomes visible with the survival probability recovered during the interval, divided by its current masking probability:

$$\frac{\alpha_{s,j} - \alpha_{t,j}}{1 - \alpha_{t,j}} = \frac{t^{c_j} - s^{c_j}}{t^{c_j}} = 1 - \left(\frac{s}{t}\right)^{c_j}.$$

With $t = 1$ and $s = 1/2$, this is $1 - 2^{-c_j}$:

$$\left(1 - 2^{-8},\ 1 - 2^{-4},\ 1 - 2^{-2},\ 1 - 2^{-1}\right) = \left(\frac{255}{256},\ \frac{15}{16},\ \frac{3}{4},\ \frac{1}{2}\right).$$

Therefore, Option B. After the first step, offset 1 is almost surely visible while offset 4 is visible half the time, so we tend to fill each block from the left. With $\kappa = 1$, every entry would be $1/2$, as in the code from Lecture 4, where a masked position is revealed with probability $(t - s)/t$.

Option A lists the offsets in reverse order. Option C lists the probabilities of staying masked. Option D is the linear schedule, which gives every offset the reveal probability $1/2$. Option E takes a first-order step $h\, c_j/t$ with $h = 1/2$ and caps each entry at one. Option F normalizes the exponents, as if exactly one offset were revealed.

**(b)** Later in generation, we can save model calls with first-order $\tau$-leaping. However, every probability in a step must stay valid.

At $t = 1/2$, all four offsets of a block are still masked. In a first-order step of length $h$, offset $j$ becomes token $a$ with probability $h\, \hat{R}_t^\theta(m \to a)$ and stays masked with the remaining probability. What is the largest $h$ for which every probability in this step is valid? Choose ONE.

A. $1/2$

B. $1/30$

C. $1/8$

D. $1/4$

E. $1/64$

F. $1/16$

(0.5 points)

The total reverse rate at a masked offset $j$ sums the model rates over the tokens:

$$\sum_{a \in \mathcal{V}} \frac{c_j}{t}\, d_\theta^j(a \mid z, x^{<b}) = \frac{c_j}{t}.$$

At $t = 1/2$, these totals are 16, 8, 4, and 2 for offsets 1 to 4. In Lecture 4, we required $h$ times every position's total rate to be at most one, so that the probability of staying masked, $1 - h\, c_j/t$, is not negative. The offsets update independently, so the condition applies to each offset, and the largest total rate determines the maximum step size:

$$16\, h \le 1 \quad \Longrightarrow \quad h \le \frac{1}{16}.$$

Therefore, Option F. Staggering the masking schedules tightens this limit early in generation, since offset 1 is revealed eight times as fast as offset 4.

Option A uses the smallest total rate, from offset 4. Option B sums the totals over the four offsets, although each offset has its own probability of staying masked. Option C omits the division by $t$. Option D multiplies $c_1$ by $t$. Option E multiplies the largest total by the number of offsets.

**(c)** In P2, we revisit earlier choices using a corrector at a fixed time. Under its rate matrix $C_t = R_t + \hat{R}_t$, the marginal $p_t$ remains stationary.

Write $d$ for $d_\theta^j(a \mid z, x^{<b})$. At time $t$, which pair gives the rate of remasking a visible token at offset $j$ and the rate of revealing token $a$ at a masked offset $j$ under $C_t$? Choose ONE.

A. $\big(\beta_j(t),\ \ \frac{c_j}{t}\, d\big)$

B. $\big(\frac{c_j}{t},\ \ \beta_j(t)\, d\big)$

C. $\big(\beta_j(t),\ \ \beta_j(t)\, d\big)$

D. $\big(\frac{c_j}{t},\ \ \frac{c_j}{t}\, d\big)$

E. $\big(0,\ \ \frac{c_j}{t}\, d\big)$

F. $\big(\beta_j(t) + \frac{c_j}{t},\ \ \frac{c_j}{t}\, d\big)$

(0.5 points)

The corrector adds the forward and reverse rate matrices at the same time $t$. A visible token $a$ at offset $j$ can jump to $m$ only through the forward rate, since the reverse process never masks:

$$C_t(a, m) = R_t(a, m) + \hat{R}_t(a, m) = \beta_j(t) + 0.$$

A masked offset can become $a$ only through the reverse rate, since the forward process never unmasks:

$$C_t(m, a) = R_t(m, a) + \hat{R}_t(m, a) = 0 + \frac{c_j}{t}\, d.$$

Therefore, Option A. By the reversal identity, $p_t(x)\, C_t(x, y) = p_t(y)\, C_t(y, x)$, so $p_t$ stays stationary under $C_t$. At $t = 1/2$, the remasking rates are $\beta_1 = 16/255$ and $\beta_4 = 2$, so offset 1 is rarely remasked and offset 4 is often remasked, in the same proportions as in the forward process.

Option B swaps the two rates. Option C uses the forward rate for revealing, which belongs to the reverse direction. Option D uses the reveal weight for remasking. Option E drops the forward part, which leaves an ordinary reverse step. Option F adds the reveal weight to the remasking rate, although the reverse process never masks.

**(d)** Each reverse step costs one network call, since we cache the clean prefix. Fewer steps per block therefore save time.

Suppose we take a single exact posterior step per block, from $t = 1$ to $s = 0$, and write $m^{L'}$ for the all-mask block. From which distribution do we draw the block $x^b$? Choose ONE.

A. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid x_1^b, \ldots, x_{j-1}^b, x^{<b}\big)$, each offset given the offsets before it

B. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid x_{-j}^b, x^{<b}\big)$, each offset given the other offsets of the block

C. $d_\theta^1\big(x_1^b \mid m^{L'}, x^{<b}\big)$ at offset 1, with the other offsets still masked

D. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid m^{L'}, x^{<b}\big)$, each offset given only the prefix

E. The data distribution of the block given the prefix, since every reveal probability is correct

F. Each offset becomes visible with probability $1 - 2^{-c_j}$ and is drawn from $d_\theta^j(\cdot \mid m^{L'}, x^{<b})$

(0.5 points)

In an exact posterior step from $t$ to $s$, each masked offset is updated independently given the current block. It becomes visible with probability $1 - (s/t)^{c_j}$, and its token is then drawn from $d_\theta^j(\cdot \mid z, x^{<b})$. With $t = 1$ and $s = 0$, every reveal probability equals one, and the current block is $m^{L'}$. We therefore draw all four tokens at once,

$$\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid m^{L'}, x^{<b}\big),$$

each from its own prediction given only the prefix. Therefore, Option D. The four tokens cannot depend on one another, so the block can join the start of one phrase to the end of another. In Lecture 4, we noted that larger steps save model calls, but the rates become outdated once another position changes. With many small steps, each reveal is conditioned on the tokens already revealed, so we can capture dependence between tokens.

Option A is the left-to-right chain, which we approach only with many steps and large $\kappa$. Option B conditions each offset on tokens that have not been generated. Option C draws only offset 1, although every reveal probability equals one when $s = 0$. Option E is false, because the step draws each offset from its own marginal, and the product of these marginals differs from the joint distribution of the block. Option F uses the reveal probabilities of the step that ends at $s = 1/2$.

## Step 5: Guiding Generation toward a Topic (2 points)

So far, we have generated LM1B sentences without controlling their topic. We now want to generate sentences about a chosen topic without retraining the denoiser. As in Lecture 4, we can condition the reverse process on an event $E$ with Doob's $h$-transform. Here $E$ is the event that a topic classifier labels the finished sentence as sports (Figure 4).

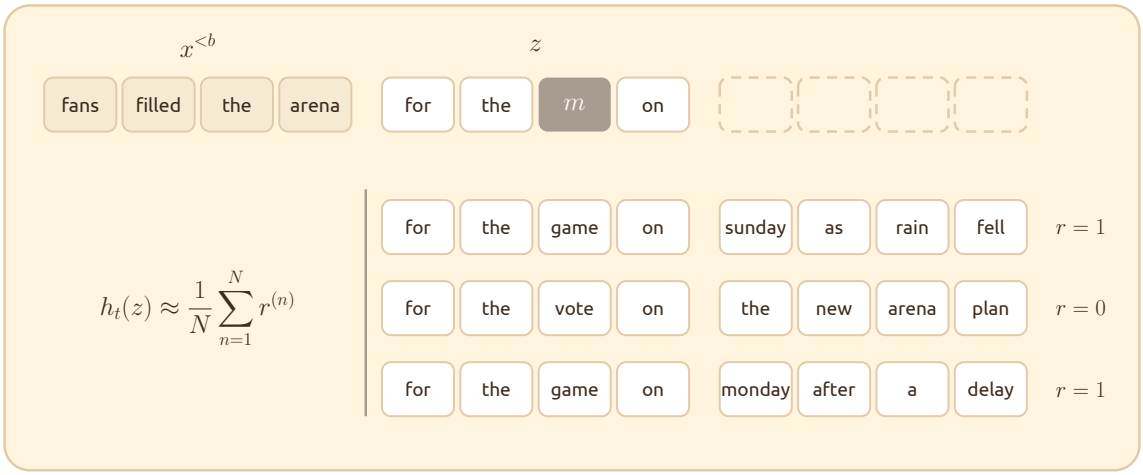



**Figure 4.** Guiding one block toward a topic. The event $E$ depends on the finished sentence, so the value $h_t(z) = \Pr(E \mid Z_t = z, x^{<b})$ averages over the masked offsets of the current block and over every later block. We estimate it by completing the sentence with rollouts and scoring each completion.

**(a)** The event concerns the finished sentence, and we generate that sentence one block at a time.

Let $h_t(z) = \Pr(E \mid Z_t = z, x^{<b})$ for the current block, and write $d$ for $d_\theta^j(a \mid z, x^{<b})$. Which expression is the guided rate of revealing token $a$ at a masked offset $j$? Choose ONE.

A. $\frac{c_j}{t}\, d\, \frac{h_t(z)}{h_t(z^{j \leftarrow a})}$

B. $\frac{c_j}{t}\, d\, h_t(z^{j \leftarrow a})$

C. $\frac{c_j}{t}\, d\, \frac{h_t(z^{j \leftarrow a})}{h_t(z)}$

D. $\beta_j(t)\, d\, \frac{h_t(z)}{h_t(z^{j \leftarrow a})}$

E. $\frac{c_j}{t}\, d\, \frac{\Pr(E \mid x^{<b})}{h_t(z)}$

F. $\frac{c_j}{t}\, d + h_t(z^{j \leftarrow a}) - h_t(z)$

(0.5 points)

Doob's $h$-transform multiplies each rate by the value at the destination divided by the value at the origin, $R^h(x, y) = R(x, y)\, h(y)/h(x)$. The unguided rate of revealing $a$ at offset $j$ is $\frac{c_j}{t}\, d$, its origin is $z$, and its destination is $z^{j \leftarrow a}$, so

$$\hat{R}_t^h\left(z, z^{j \leftarrow a}\right) = \frac{c_j}{t}\, d\, \frac{h_t(z^{j \leftarrow a})}{h_t(z)}.$$

Therefore, Option C. The value conditions on the prefix and the noisy block, and it averages over the masked offsets and every later block, because $E$ depends on the finished sentence. We increase the rates of reveals that make $E$ more likely and decrease those that make it less likely. The denoiser stays fixed.

Option A inverts the ratio, which would steer away from the topic. Option B omits the division by $h_t(z)$, so the rates would shrink with the overall value of the block. Option D multiplies the forward rate $\beta_j(t)$, which belongs to masking, and it also inverts the value ratio. Option E uses the value of the prefix alone at the destination. Option F adds the change in value to the rate, although the $h$-transform multiplies the rate by the ratio of values.

**(b)** At $t = 1/2$, offset 3 of the current block is masked, and $c_3 = 2$.

The denoiser assigns probabilities 1/2, 1/4, and 1/4 on the words *game*, *vote*, and *film*, and zero probability to all other words. A classifier gives $h_t(z) = 0.4$ for the current block, and $h_t(z^{3\leftarrow a}) = 0.6$, 0.2, and 0.2 after revealing *game*, *vote*, and *film*. Which guided rates toward (*game*, *vote*, *film*) does the $h$-transform give? Choose ONE.

A. (2, 1, 1)

B. (4/3, 2, 2)

C. (6/5, 1/5, 1/5)

D. (3/4, 1/8, 1/8)

E. (3, 1/2, 1/2)

F. (12/5, 4/5, 4/5)

(0.5 points)

At $t = 1/2$ with $c_3 = 2$, the weight is $c_3/t = 4$, so the unguided rates are $4 \cdot (1/2, 1/4, 1/4) = (2, 1, 1)$. The value ratios are

$$\frac{h_t(z^{3\leftarrow a})}{h_t(z)} = \left(\frac{0.6}{0.4}, \frac{0.2}{0.4}, \frac{0.2}{0.4}\right) = \left(\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

Multiplying gives the guided rates (3, 1/2, 1/2). Therefore, Option E. The rate toward *game* rises by half, and the rates toward *vote* and *film* halve. The total stays at 4, because the denoiser-weighted average of the values after the reveal equals the current value, $\frac{1}{2}(0.6)+\frac{1}{4}(0.2)+\frac{1}{4}(0.2) = 0.4$. With an exact classifier, guidance changes which word appears at offset 3, and the offset still becomes visible at its unguided total rate.

Option A is unguided. Option B inverts the ratios. Option C multiplies by the value after the reveal without dividing by $h_t(z) = 0.4$. Option D is the guided distribution over the three words, the rates divided by their total of 4. Option F multiplies the weight by the value after the reveal and drops the denoiser.

**(c)** A classifier on noisy blocks can be hard to train. In PepTune, the value is estimated by completing the sequence with the denoiser and averaging the rewards of the completions.

We estimate the values at offset 3 the same way. Each completion fills the rest of the current block and every later block, and it earns reward 1 if the classifier labels the finished sentence as sports and 0 otherwise. The completions after revealing *game* earn $(1, 1, 0, 1)$, and those after revealing *vote* earn $(0, 1, 0, 0)$. The current value estimate is 0.5, and the unguided rates toward *game* and *vote* are 2 and 1. Given that the next reveal at offset 3 is *game* or *vote*, what is the probability that it is *game*? Choose ONE.

A. 2/3

B. 3/4

C. 2/5

D. 11/16

E. 3/5

F. 6/7

(0.5 points)

Each value estimate is the average reward of the completions:

$$h(game) = \frac{1+1+0+1}{4} = \frac{3}{4}, \qquad h(vote) = \frac{0+1+0+0}{4} = \frac{1}{4}.$$

The $h$-transform multiplies each unguided rate by its value ratio:

$$2 \cdot \frac{3/4}{1/2} = 3, \qquad 1 \cdot \frac{1/4}{1/2} = \frac{1}{2}.$$

Given that the next reveal is one of the two words, its probabilities are proportional to these rates:

$$\frac{3}{3+1/2} = \frac{6}{7}.$$

Therefore, Option F. The current value of 0.5 cancels from this ratio. The completions must fill the later blocks too, because the classifier scores the finished sentence. Without guidance, *game* would come first with probability 2/3.

Option A is the unguided probability. Option B compares the two values without the base rates. Option C inverts the value ratios. Option D adds each value to its rate. Option E swaps the base rates of the two words.

**(d)** We repeat rollouts for every sample. In TR2-D2, we retain the search results and use them to fine-tune the model toward a reward-tilted target. We can then sample without rollouts.

Two completions of a prefix have model probabilities 3/4 for a sentence about politics and 1/4 for a sentence about sports, with rewards 0 and $\log 3$. The target is $p^\star(x) \propto p_\theta(x)\exp\big(r(x)/\lambda\big)$. Which target probabilities of (politics, sports) does temperature $\lambda = 1/2$ give? Choose ONE.

A. (1/2, 1/2)

B. (1/4, 3/4)

C. (3/4, 1/4)

D. (27/28, 1/28)

E. (1/10, 9/10)

F. (0, 1)

(0.5 points)

The tilted target multiplies each model probability by $\exp(r/\lambda)$. At $\lambda = 1/2$, this factor is $\exp(2r)$, so the unnormalized weights are

$$\text{politics: } \tfrac{3}{4}\,e^{0} = \tfrac{3}{4}, \qquad \text{sports: } \tfrac{1}{4}\,e^{2\log 3} = \tfrac{1}{4}\cdot 9 = \tfrac{9}{4}.$$

The weights sum to 3, so the target probabilities are (1/4, 3/4). Therefore, Option B. At $\lambda = 1$, the sports weight would be 3/4 and the target would be (1/2, 1/2). As in Lecture 4, lowering the temperature increases the probability assigned to the higher-reward sentence. This target is the $h$-transform with $h = \exp(r/\lambda)$ at the end of generation, and after fine-tuning we sample it without rollouts.

Option A uses $\lambda = 1$. Option C applies no tilt. Option D tilts with $\exp(-r/\lambda)$, which favors the sentence without reward. Option E drops the model probabilities. Option F multiplies by $r/\lambda$, which is zero for the politics sentence.

## Step 6: Interpreting Hypothetical LM1B Results (2 points)

We trained staggered block diffusion on LM1B with $L' = 4$ at the BD3-LM budget, for several values of $\kappa$. We will interpret the hypothetical results in Table 1 and Figure 5 using the facts derived above.

| Model | $\kappa$ | Training perplexity | Test perplexity |
|---|---|---|---|
| Transformer | – | 18.42 | 22.83 |
| BD3-LM | 1 | $\leq$ 27.41 | $\leq$ 28.23 |
| Staggered | 2 | $\leq$ 22.90 | $\leq$ 24.10 |
| Staggered | 4 | $\leq$ 20.31 | $\leq$ 21.95 |
| Staggered | 16 | $\leq$ 19.07 | $\leq$ 22.41 |
| Staggered | 256 | $\leq$ 18.61 | $\leq$ 22.79 |

**Table 1.** Hypothetical LM1B perplexities at the BD3-LM budget with blocks of four. The transformer's values are exact, and the diffusion values are bounds.

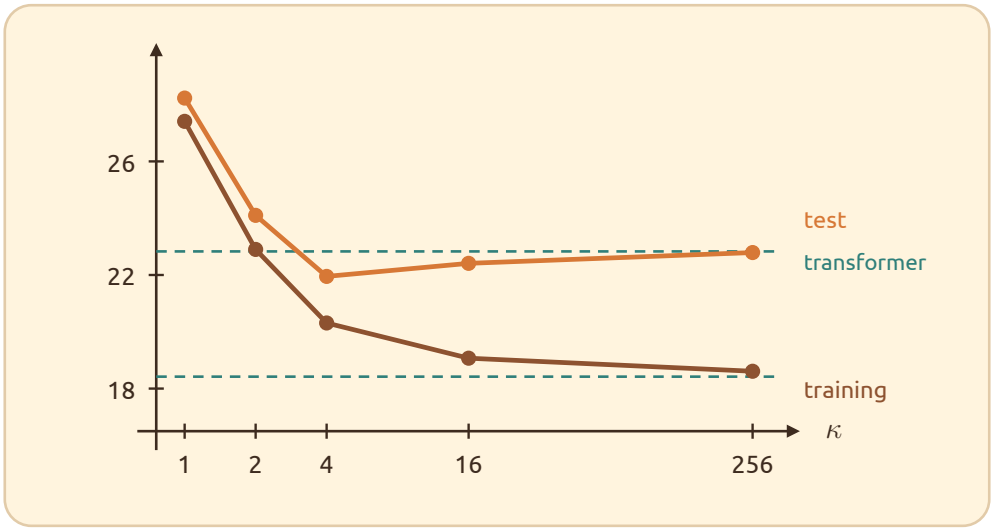



**Figure 5.** The values of Table 1 against $\kappa$ on a logarithmic axis. Dashed lines mark the transformer's training and test perplexities.

**(a)** With $\kappa = 4$, we reached the lowest test value in Table 1.

Which conclusion does its bound of 21.95 support, compared with the transformer's 22.83? Choose ONE.

A. Its test perplexity is at most 21.95, so it is below the transformer's 22.83.

B. Nothing follows about the comparison, because a bound on perplexity may be loose.

C. Its test perplexity is at least 21.95, so it may still exceed 22.83.

D. It beats the transformer only if its bound turns out to be tight.

E. A bound and an exact perplexity cannot be compared in either direction.

F. Its test perplexity is exactly 21.95, because the staggered bound is tight.

(0.5 points)

The reported value bounds the negative log-likelihood, $-\log p_\theta(x) \le \mathcal{L}(x)$, and exponentiating the averages per token preserves the inequality. The test perplexity of the $\kappa = 4$ model is therefore at most 21.95, which is below the transformer's exact 22.83. Therefore, Option A. In Lecture 4, we noted that a low bound forces a low negative log-likelihood, while a high bound may be loose. In 1(a), the bound was above 22.83, so we could not rank the two models. Here the bound is below 22.83, which settles the comparison.

Option B applies the caution about loose bounds to a bound below the transformer. Option C reverses the inequality. Option D incorrectly requires the bound to be tight for this conclusion to hold. Option E is false, because an upper bound below an exact value settles the comparison. Option F is false, because the rate loss does not guarantee a tight bound.

**(b)** In Table 1, we also list each model's perplexity on its training sentences.

Which interpretation of the $\kappa$ sweep agrees with both columns? Choose ONE.

A. The bound loosens at every offset as $\kappa$ grows, so the training and test values both rise with $\kappa$.

B. Larger $\kappa$ adds a positive prior term at $t = 1$, which raises the test value and leaves training unchanged.

C. Larger $\kappa$ raises the weights $c_j/t$, so we reach a lower loss on the training sentences and the test sentences alike.

D. Beyond $\kappa = 4$, the training value keeps falling while the test value rises, which is the pattern of overfitting.

E. With larger $\kappa$, we need fewer network calls, which lowers the training value and raises the test value.

F. The test value must fall as $\kappa$ grows, so its rise after $\kappa = 4$ reflects noise in the evaluation.

(0.5 points)

Reading down the training column, the training value falls from 27.41 at $\kappa = 1$ to 18.61 at $\kappa = 256$,

close to the transformer's 18.42. The test value is lowest at $\kappa = 4$ and rises toward the transformer's 22.83 as $\kappa$ grows. The gap between the test and training values widens with $\kappa$,

$$0.82, \quad 1.20, \quad 1.64, \quad 3.34, \quad 4.18, \qquad \text{against } 4.41 \text{ for the transformer.}$$

Beyond $\kappa = 4$, we fit the training sentences better without fitting new sentences better, which is overfitting. Both columns contain bounds, so this interpretation assumes that the gap between each bound and the true perplexity changes little across $\kappa$. Therefore, Option D. As $\kappa$ grows, the order of masking inside each block becomes nearly the same every time, since later offsets are almost always masked first. Training then approaches the single left-to-right task of the transformer. At $\kappa = 4$, we still train on varied corruptions while generating each block nearly from left to right, so we would choose $\kappa$ on held-out text.

Option A contradicts the training column, which falls. Option B is false, because every offset ends at the mask for every $\kappa$, so the prior term is zero. Option C contradicts the test column, which rises after $\kappa = 4$. Option E attributes the trend to network calls, which depend only on the number of sampling steps and enter neither column. Option F requires a monotone trend that nothing in the model implies.

**(c)** In Run E, we repeated the run with $\kappa = 4$, using MDLM's weight $1/t$ for every masked offset in the loss. The test value computed with this weight was 2.8.

What can we conclude from this number? Choose ONE.

A. It is a valid bound and a tighter one, because $1/t \le c_j/t$ at every offset.

B. It is a valid but looser bound, because the weight $1/t$ is the same at every offset.

C. It may not bound the test perplexity, because $1/t \le c_j/t$ and the smaller weight shrinks the masked terms.

D. It is the exact test perplexity, because $1/t$ is the reveal weight of the linear schedule.

E. It is a valid bound, because the forward process still ends at the all-mask block.

F. It may not bound the test perplexity, because $1/t \ge c_j/t$ and the larger weight inflates the masked terms.

(0.5 points)

Our bound came from the rate error with model rates $\frac{c_j}{t}\,d$, which produced the weight $c_j/t$ in 3(b). In Run E, we computed a different number. Every $c_j \ge 1$, so the weight $1/t$ is at most $c_j/t$ at every offset, and each masked term shrank. At $\kappa = 4$, the exponents are $(64, 16, 4, 1)$, so the term at offset 1 shrank 64-fold. Nothing guarantees that the smaller number stays above the negative log-likelihood. Therefore, Option C. Integrating against the masking probability shows the size of the change:

$$\int_0^1 \frac{1}{t}\, t^{c_j}\, \mathrm{d}t = \frac{1}{c_j}, \qquad \int_0^1 \frac{c_j}{t}\, t^{c_j}\, \mathrm{d}t = 1,$$

so offset $j$ has total weight $1/c_j$ in Run E and 1 in our loss. If the four offsets contribute equally to our bound of 21.95, the value in Run E is

$$\exp\Big(\tfrac{1}{4}\big(\tfrac{1}{64} + \tfrac{1}{16} + \tfrac{1}{4} + 1\big) \log 21.95\Big) \approx 2.8,$$

which matches the value we obtained. Every LM1B model in Table 1 lies far above 2.8, so we should suspect an error in the loss.

Option A would first require a proof that the smaller number bounds the likelihood at all. Option B is false, because a number below our bound cannot be a looser bound. Option D is false even for the linear schedule, where the weight $1/t$ still gives only a bound. Option E rests on the prior term, which is independent of the weights. Option F reverses the comparison of the two weights.

**(d)** In Run F, we repeated the run with $\kappa = 4$ and also supplied the time $t$ to the denoiser, as in the code from Lecture 4. We obtained the same test bound of 21.95.

Which fact explains why adding the time input did not improve the bound? Choose ONE.

A. The weight $c_j/t$ already brings $t$ into the loss, so a time input to the network is redundant.

B. At a given time, every offset of the block has the same masking probability, which removes the stagger from the posterior.

C. The prefix is clean at every time, so time has no effect on the denoiser's prediction.

D. By Bayes' rule, the posterior depends on $t$ through $\alpha_{t,j}$, and the number of masks in $z$ determines $t$.

E. In Bayes' rule, the factors $\alpha_{t,j}$ and $1 - \alpha_{t,j}$ are the same for every clean block that agrees with $z$, so they cancel.

F. The weights $c_j/t$ are the same at every offset, so the time cancels from the loss.

(0.5 points)

Bayes' rule gives the posterior over the clean block,

$$\Pr\big(x^b \mid Z_t = z, x^{<b}\big) \propto p\big(x^b \mid x^{<b}\big) \prod_{j \text{ visible}} \alpha_{t,j}\, \delta(z_j, x^b_j) \prod_{j \text{ masked}} \big(1 - \alpha_{t,j}\big).$$

For every clean block that agrees with the visible tokens of $z$, the product of the factors $\alpha_{t,j}$ and $1 - \alpha_{t,j}$ is the same, so it cancels in the normalization. At every $t$, the posterior is the data distribution of the masked offsets given the visible ones and the prefix. Therefore, Option E. The optimal denoiser is the same at every $t$, which is why we wrote $d^j_\theta(a \mid z, x^{<b})$ without $t$, as in the BD3-LM loss of Lecture 4. In Run F, the optimal denoiser was the same with or without the time input, so the bound stayed at 21.95.

Option A is false, because the loss weight does not enter the posterior. Option B is false for $\kappa > 1$, since the exponents differ across offsets. Option C holds for the prefix, but the question concerns the noisy block. Option D is false on both counts, because the time dependence cancels and many times give the same number of masks. Option F is false, because the weights $c_j/t$ differ across offsets, and the loss weights do not enter the posterior.

**End of practice exam.**

*This page is left intentionally blank for scratch work*